

Phanidhar Akula

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EDUCATION

Miami University

Aug. 2024 – Aug. 2026

M.S. Computer Science, GPA: 3.86/4.00

- **Leadership:** President, Graduate Students of Color Association | Elected Graduate Student Representative, University Graduate Council | Program Lead, NAE Grand Challenges Scholars Program

Malla Reddy University

Aug. 2020 – May 2024

B.Tech Computer Science

TECHNICAL SKILLS

Python, TypeScript, JavaScript, Java, C++, SQL, ML, GenAI, LLMs, RAG, React, Next.js, Redux, Tailwind, Node.js, FastAPI, REST APIs, OAuth, PostgreSQL, MySQL, Supabase, AWS, Docker, CI/CD, Git, Linux, Vercel, iOS, Android

EXPERIENCE

Computer Science Researcher

Aug. 2024 – Jul. 2026

Miami University

- Developed SimForge, an open-source Python framework for urban traffic simulation that benchmarks 3 simulators under identical conditions from one input, with Docker, CI, and 668 pytest tests for full reproducibility.
- Built high-performance parallel routing models for transportation simulation, caching per-resident routes and parallelizing across 16 supercomputer cores in Python to cut runtime from 142 hours to 7 (20x speedup).
- Debugged Cityscape, an open-source C++ digital twin for commute modeling, tracing a geospatial data-loading bug to one filter; 3 merged PRs restored 493K buildings and cut invented-home placements from 70% to 6%.
- Designed a 16-configuration Python modeling study across 5 cities on a supercomputer cluster, tuning router search-radius parameters and improving worst-case commute-time prediction accuracy from 81% to 92%.

Software Engineer Intern

Jan. 2024 – May 2024

Smart Pathfinders Overseas Education

- Shipped a private client portal in React and Redux, giving 180+ enrolled students one centralized place to track applications, upload documents, and reach counselors directly instead of emailing staff for updates.
- Implemented Python/MySQL REST APIs and data models behind role-based access control across 3 user roles, so each student saw only their own records while staff managed the full caseload from one central dashboard.

Full-Stack Developer

May 2023 – Dec. 2023

Freelance – E-Commerce & Fitness Startup Clients

- Delivered production-grade web applications for clients, building React/Redux interfaces against Python/MySQL REST APIs and deploying to AWS, replacing manual spreadsheet workflows for 300+ daily users.
- Migrated 10K+ records from client Excel workbooks into MySQL with a custom Python ETL script, validating row counts and key fields against the source so both clients cut over in one pass with no manual re-entry.

PROJECTS

LumiAI | AI Tutoring Platform, studywithlumi.com

Oct. 2025 – Present

- Built and shipped a production-grade React/TypeScript platform spanning document-grounded chat, quizzes, flashcards, spaced repetition, and voice tutoring, serving 220+ students and handling 3K+ AI interactions.
- Engineered a serverless LLM inference layer on Vercel with token-verified requests, response streaming, and prompt caching, blocking 100% of anonymous inference calls and serving 87% of input tokens from the cache.
- Architected the full-stack data layer on PostgreSQL with database-enforced row-level security across all 13 user data tables, Google OAuth login, and signed storage, so no query path can return another user's data.
- Extended the platform to iOS and Android using Capacitor, reusing 95% of the existing TypeScript codebase in native shells with platform-specific auth and storage and submitting both builds for app store review.

PUBLICATIONS (UNDER REVIEW, 2026)

Akula, P., Rao, D. M. *Attributable Cross-Simulator Benchmarking for Urban Traffic Simulation.* IEEE T-ITS.

Akula, P., et al. *CITYSCAPE: A Digital Twin Framework for Commute Dynamics.* Frontiers (Revision submitted).